

Defining Embodied Provenance for Immersive Sensemaking

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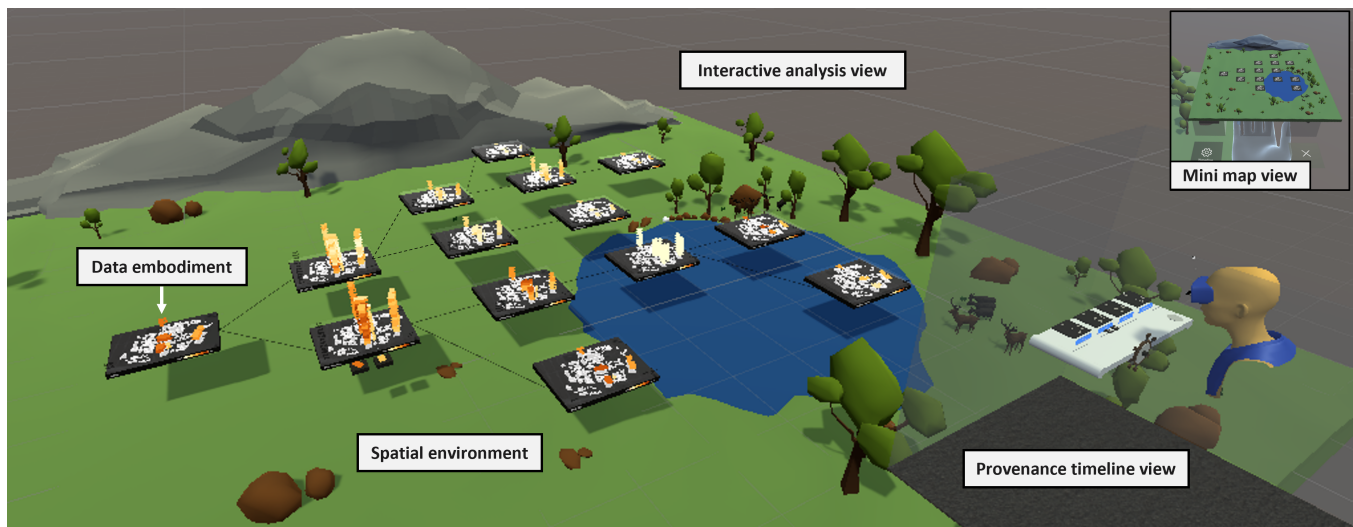


Figure 1: We explore the concept of embodied provenance with a prototype implementation that allows users to interact with embodied data representations in a large spatial environment. Users can conduct complex analyses of spatio-temporal data in the interactive analysis view with the help of a mini map view for navigation and overview. In this figure we see a user in the control booth above, observing a previous analysis using the provenance timeline view.

ABSTRACT

Immersive analytics research has explored how embodied data representations and interactions can be used to engage users in sensemaking. Prior research has broadly overlooked the potential of immersive space for supporting analytic provenance, the understanding of sensemaking processes through users' interaction histories. We propose a concept of embodied provenance, the use of three-dimensional space and embodiment in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We highlight a set of design criteria for analytic provenance drawn from prior work and propose a conceptual framework for embodied provenance. We develop a prototype system in

virtual reality to demonstrate the concept and support the conceptual framework by providing multiple data views and embodied interaction metaphors in a large space. We present a use case scenario of energy consumption analysis, which shows the system's potential for assisting analytic provenance using embodiment.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality; Interaction techniques; Visualization.**

KEYWORDS

embodied provenance, immersive analytics, sensemaking, analytic provenance, spatial memory

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1 INTRODUCTION AND RELATED WORK

Immersive analytics [4, 18] is an emerging field that builds upon the fields of data visualization, visual analytics, virtual reality and augmented reality, and human-computer interaction, which extends traditional desktop visualization into a variety of immersive environments. It has brought rich opportunities for the exploration of data visualisations through engaging embodied data constructs and interactions. Embodiment has been used in recent research to support sensemaking, by representing abstract data as virtual objects to enable direct manipulation and intuitive interactions. For example, ImAxes [5] allows users to construct ad-hoc visualisations by combining individual axes, while TimeTables [31] allows users to ‘drill down’ into temporal data by reaching into the data structure to pull out layers and stretch their time dimension. Recent works have also taken advantage of large spatial environments afforded by Virtual Reality (VR) for organising and browsing embodied data constructs (e.g. DataHop [11], TimeTables [31]), to support sensemaking.

Sensemaking is the process of extracting information and building understanding from a collection and organisation of data [1, 15]. More recently, researchers have come to realise that collecting and reflecting on *analytic provenance* information — knowledge about the history of users’ interactions and reasoning process [30] — is in itself an important part of sensemaking [26, 28]. Analytic provenance can be used for various purposes [21], such as recalling the analysis process [27], assisting narrative construction for outcome presentation [10, 20], reproducing analysis workflow [7, 24], and supporting collaborative sensemaking and communication by sharing the analysis process [6, 19]. Various studies have shown the value of interaction histories and their benefits in information sensemaking [26, 28].

To date, there has been little exploration of supporting provenance in immersive analytics applications. Immersive Space to Think [15] enables note taking to support external memory for insights when performing sensemaking tasks, but the features are primarily focused on supporting the data specification and view manipulation tasks. Spacetime [29] allows replay motion and view sharing, but it emphasises on novel scene editing features during the analysis process rather than the interaction histories. To our knowledge, there has not been any systematic attempt for immersive analytics provenance using spatial organisation of embodied data constructs and interaction metaphors.

Building on the achievements of immersive analytics, we believe that embodiment and immersive space can be used for supporting analytic provenance. Previous research in immersive analytics found that embodiment intervention and spatial distributions reduce cognitive load and shows promises to aid spatial memory [2, 14]. Large volumes of provenance data may be generated when conducting complex analysis tasks, and the use of embodiment could help to reduce users’ cognitive load [13, 14] by externalising the data into the spatial environment.

In this paper, we propose the concept of *embodied provenance*, which we define as the use of three-dimensional space and embodiment in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We highlighted a full set of criteria drawn from prior work on analytic provenance and

established a conceptual framework by identifying three key properties for embodied provenance. We further implemented a prototype system and presented a use case scenario to demonstrate the concept in VR, by providing multiple data views, spatial environments and novel physical interactions. This research sheds light on the gaps of how to leverage the benefits of embodied data constructs and interactions to improve the understanding of analysis history to support sensemaking.

2 EMBODIED PROVENANCE

To enable the future exploration of systems that support immersive sensemaking, we aim to more firmly establish our motivation and definition of embodied provenance. First, we outline the design criteria for analytic provenance.

2.1 Design Criteria for Analytic Provenance

Through our review of the literature we identify three prominent theoretical work on analytic provenance from which provenance criteria are commonly cited. Ragan et al.[21] organised purposes of provenance, Cutler et al.[6] presented several design goals based on these purposes and Stuerzlinger et al.[25] defines provenance tasks in immersive analytics. To define our five key design criteria for analytic provenance, we select the main overlapping themes from these works:

C1. Support Memory Recovery: The system should support users to recall their previous analysis. This recovery of process memory should enable awareness and understanding of what analytic tasks have previously been completed, what findings have been established, and what tasks remain to be completed in the future [21, 25].

C2. Support Action Recovery: The system should support users to maintain the action history during the analysis process. Users should have the freedom to conduct exploratory analysis and the ability to correct the human errors and return to previous state of analysis through action recovery [6, 21, 25].

C3. Support Replication and Reproducibility: The system should support users to replicate and reproduce the steps or workflow of a previous analysis. Replication pertains to the analysis, whereas reproducibility is primarily concerned with the ability to obtain consistent results of the analysis. For instance, users should be able to conduct branching investigations or revise the analysis without modifying the previous states [6, 21].

C4. Support Annotation: The system should allow for a means of note taking during the analytical process and for post-analysis activities. This could include techniques such as highlighting visualisation, adding text comment, or voice memo. These annotations should enable users to keep track of significant insights in the context in which they were found and to communicate their analysis to others [6, 21, 25].

C5. Support Sharing and Collaboration: The system should enable users to communicate and share insights in synchronous or asynchronous collaborative analysis settings. This collaborative analysis should allow for information sharing such as provenance history or annotations [6, 21, 25]. Furthermore, the sense of presence of others is also an important factor in embodied provenance, which can be achieved by showing avatars of other users.

2.2 Embodying Analytic Provenance

Given the above collection of design criteria, the central question of this research is how to best support analytic provenance in immersive analytics applications.

We draw inspiration from previous work on spatial interaction and embodied interaction. For instance, DataHop [11] takes advantage of a large environment afforded by VR to encode interaction history as three-dimensional, room-sized graph structure. Conversely, ImAxes [5] uses direct manipulation of embodied graph axis for intuitive interaction with high-dimensional data. TimeTables [31] combines both approaches, using direct manipulation of space-time cube elements to create persistent views that capture the analysis history. Building on this work, we propose a conceptual framework for embodied provenance by describing the properties needed for a system to provide such support.

P1. Use embodied representation of data construct and interaction metaphor. Embodied representation of both data constructs and interaction metaphors is crucial in embodied provenance. Instead of visualising the analysis history as abstract representations (e.g., node-link diagram), embodied provenance approach would represent events as embodied three-dimensional representations. Embodied provenance also includes the use of visual metaphors of physical objects to represent the system's components and interactions.

This property fosters intuitive and fluid interaction [8] with embodied data elements. For example, TimeTables scenario [31] integrates the selection and manipulation of the data layer with the creation of a 'zoomed-in' view of the layer in the TimeTables. Further, the persistent tables created act as nodes in a resulting provenance graph.

P2. Seamless integration between temporal and spatial aspects. Temporal aspect (e.g., history of events) and spatial aspects (e.g., representations of this history) are important factors in analytic provenance. Traditionally, the amount of temporal information presented is subject to the display space budget, meaning the layout and type of event history representation is constrained to a limited 2D space. This is not the case for embodied provenance environments with vast display spaces. Embodied provenance make use of this space to visualise great amount of historical information. For example, lay out data analysis steps as network of 3D charts [11] or virtual tabletops [31] in the large virtual environment. Integrating temporal aspect to three-dimensional space might also to support information acquisition as people generally have a better sense of their surroundings, i.e. environment awareness[12].

P3. Aid spatial memory through environmental features. As described earlier, analytic provenance processes should support memorability. Through support for strong sense of presence, embodied provenance offers a means to support this criteria that is beyond traditional analytic provenance on 2D displays. For instance, while exploring a 3D world, people gather information of their surroundings and build clues that are embedded in environmental features. Furthermore, spatial locations are integrated with events that occurred there in the process known as episodic memory formation [22] (for instance remembering where you were when a significant event occurred). Thoughtful design of recognisable environmental features can potentially support both spatial awareness

triggering of episodic memories that previously occurred in certain locations, aiding memorisation of analytic steps by recalling locations.

We will refer back to the design criteria (C1–5) and the key properties for embodied provenance (P1–3) when describing the system visualisations and interactions in the next section.

3 PROTOTYPE

We demonstrate the embodied provenance concept with a prototype system extended from the TimeTables system. As a starting point, we implemented several of the basic interactions from TimeTables. These include filtering a time layer or block by grabbing the object to make a copy, and stretching a layer or block to increase the level of time granularity, for example, from one year to monthly data.

3.1 Implementation

We built on the code made available by the authors of TimeTables. The system is built in Unity3D¹ (version 2020.3.17f1) and displayed with an Oculus Quest 2 which has hand tracking capabilities. This device is connected to a Windows 10 PC equipped with an Intel i7 processor and an NVIDIA GeForce GTX 1080 graphics card. The prototype was run at a frame rate of 72Hz. We use Mixed Reality Toolkit² (MRTK) for implementing the interactive components and hand tracking. Mapbox Unity SDK³ is installed for the campus map on the tabletop.

3.2 Visualisation

We provide various embodied data visualisations to support analytic provenance through multiple data views and large virtual environments with different themes in our system.

3.2.1 Multiple Views. The system provides multiple views to support embodied provenance inspired by Shrinivasan and van Wijk's framework [23], SenseMap [19] and the CLUE model [10]. To support analytical reasoning process, Shrinivasan and van Wijk proposed a data view for the actual visualisation of the data, a navigation view showing the exploration process in a tree, and a knowledge view to record findings with notes and link them to the visualisation states. Similarly, SenseMap includes a browser view, a history map and a knowledge map, while in the CLUE model, a provenance view and a story view are generated upon the exploration of the data visualisation. We combine and refine these concepts based on our motivation and design goals to create the following three views (See Fig. 1):

- **Interactive analysis view:** actual analysis with visualisations in a hierarchy tree structure
- **Mini map view:** an overview of the current visualisation state
- **Provenance timeline view:** analysis histories in a linear timeline structure

The interactive analysis view consists of all the visual representations and interactions for the data exploration and analysis with the surrounding environment. The visualisations are similar

¹<https://unity.com/>

²<https://developer.leapmotion.com/mrtk-unity>

³<https://www.mapbox.com/>



Figure 2: Six themed spatial environments (left) and Tool menu (right) provided by our prototype.

to those in TimeTables [31] which consists of colour encoded layers representing temporal data above virtual tabletops (**P1. Use embodied representation of data construct and interaction metaphor**), however, the layout of the tabletops in our system is a hierarchy tree as tree-like visualizations are usually used to depict the branching trace structure [9] of an analysis process. The nodes are the tabletops in different levels of time granularity (year, month, week, day and hour) and links connect between a tabletop and its child tabletop in the next detailed level of time granularity (**P2. Seamless integration between temporal and spatial aspects**). Using this layout, users can avoid the visual clutter and easily trace the relationship of different tabletops and the provenance data.

The mini map view includes a scaled and simplified interactive analysis view without the interactions for analysis. The mini map view changes in correspondence with the interactive analysis view to reflect recent changes and provides an overview of the current visualisation state. A miniature avatar indicates the user's position in the mini map view so that they can obtain a spatial awareness of where they are located in the environment. Users can also easily navigate to a chosen tabletop in the interactive analysis view by tapping on the tabletop's miniature version in the mini map view.

The provenance timeline view contains an embodied timeline to support analytic provenance tasks. A timeline representation is chosen because previous research shows that linear temporal sequences provide a clear sense of the direct history of the exploration [9]. The timeline consists of multiple tabletops, every time a new tabletop is created in the interactive analysis view, a simplified tabletop will be appended to the timeline based on the time they were created (**P2. Seamless integration between temporal and spatial aspects**). Following the metaphor of a control booth in a theatre (which provides technicians with an overview of the stage), the timeline is located above the interactive analysis view where the analyst has access to a complete view of the visualisations and exploration process in the interactive analysis view below. The timeline combines the metaphors of a conveyor belt and a ship's helm — users can rotate an embodied helm to move the timeline conveyor forward or backwards. To ensure the timeline fits easily in view, only four tabletops are shown at one time.

3.2.2 Themed Spatial Environments. In the physical world, we rely on unique features of our surrounding environments as landmarks for navigating and recalling locations. According to Lynch [16, 17], landmarks are more memorable if they have a clear form, contrast with their background, and are in a prominent spatial location. In contrast to the large empty environment in TimeTables, our system allows users to choose from six themed environments (see Fig. 2a) with spatial landmarks and audio aimed to facilitate memory recall (**C1. Support Memory Recovery and P3. Support environmental features to aid spatial memory**). Each environment has its distinct scene and background sound. Within each environment, different landmarks are presented in the space, for example, Christmas trees, snowmen, a bridge decorated with coloured lights, benches and streetlights are shown at different locations in the Christmas scene.

3.2.3 Tool Menu. We support embodied interactions using a tool metaphor [3] (**P1. Use embodied representation of data construct and interaction metaphor**). A tool menu is shown when users face the palm toward them. (see Fig. 2b). The tool menu consists of three side-by-side widgets and a button to allow quick navigation to the timeline. The most left widget contains a lever for undo and redo, the middle one includes a mini map, and the most right one has a microphone inside for annotations. Users can drag the lever, the mini map and the microphone out of the containers and these three objects will be scaled up to a easy-to-manipulate size after leaving the containers. The three interactive elements can be then put back into the containers when they are not needed.

3.3 Interactions

The system follows the interactions from TimeTables for data exploration and further provides multiple physical interactions (**P1. Use embodied representation of data construct and interaction metaphor**) in supporting embodied provenance.

3.3.1 Replay-It. The Replay-It feature allows users to replay their previous exploration processes, to help them recall what analyses have been done and the findings that were discovered (**C1. Support Memory Recovery**). The playback includes all visualisations

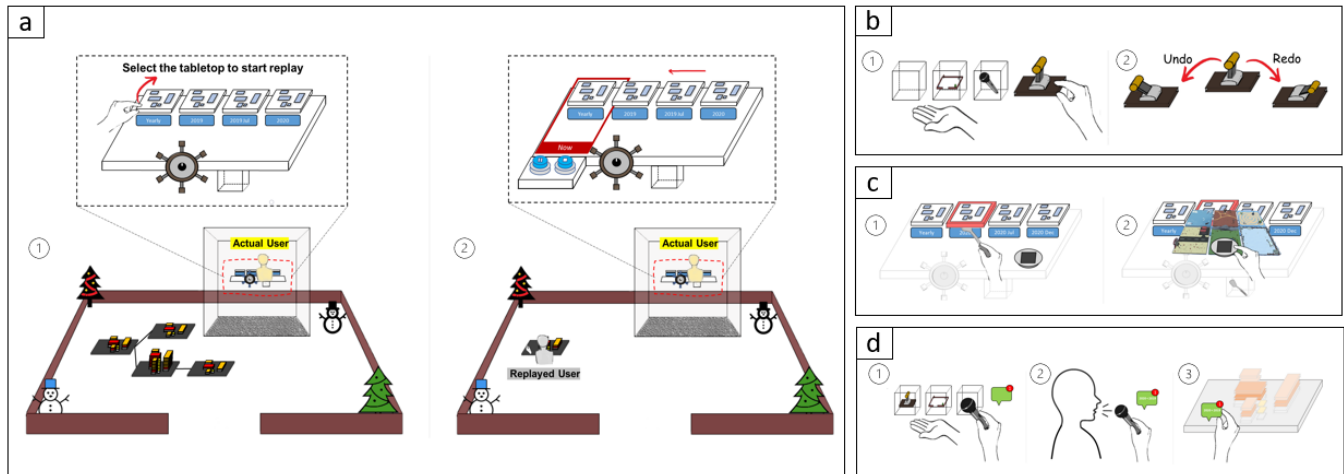


Figure 3: Interactions provided by the prototype. a) Replay-It, b) Do-It-Lever, c) Copy-It-Spatula, d) Note-Down-Mic.

created, annotations for insights, and an avatar representing the user's previous position, head movements and hands movements. In addition to analyses conducted by themselves, users are also able to watch the replay of analysis processes conducted by other analysts (**C5. Support Sharing and Collaboration**). From the timeline view, the user can start the replay by selecting the tabletop on the timeline, the system will then replay the analysis starting from the creation of the selected table in the interactive analysis view. Users can choose to stay in the control booth above the interactive analysis view to watch the replay from an overview perspective, or navigate down into the environment and look over the avatar's shoulder to get more detailed information. During the replay, the timeline will scroll along and an area highlighted with red light will indicate the current point in time. A pause button and a stop button are also provided on the timeline so that users can pause/play or stop the replay process. Fig. 3a shows how the Replay-It interaction works.

3.3.2 Do-It-Lever. We provide an embodied lever to allow undo and redo interactions which enable users to achieve the action recovery along with their analysis process (**C2. Support Action Recovery**). Users can pull a lever extracted from the tool menu to the left to undo the last action while pull the lever to the right to redo the action (see Fig. 3b). The supported actions include creating a tabletop for multiple buildings exploration, creating a pedestal with a block for the analysis of a single building, stretching a layer above the tabletop and stretching a block on the pedestal for a more detailed time granularity.

3.3.3 Copy-It-Spatula. The Copy-It-Spatula enables users to copy created tabletops and pedestals and paste them in another environment (see Fig. 3c). This feature allows users to reproduce the steps of a previous analysis process (**C3. Support Replication and Reproducibility**), or to branch off from analysis created by another user (**C5. Support Sharing and Collaboration**). When users grab the spatula from a container under the timeline, a plate for containing the copied objects will appear. Users can copy the tables or pedestals on the timeline to the plate by shoveling them

with the spatula. Users can then paste the copied visualisations into a new environment by dropping the plate onto one of the presented six environments. Users will be automatically teleported to that environment alongside the copied tabletops and pedestals. A mini version of the environment where the tabletop is copied from is created on the right corner of the tabletop, and users can navigate back to the source scene by selecting the mini environment.

3.3.4 Note-Down-Mic. The Note-Down-Mic provides users the ability to add annotations throughout the analysis process (**C4. Support Annotation**). The annotations can be ideas, comments, findings or insights. Users can add them to the analysis conducted not only by themselves, but also by other analysts (**C5. Support Sharing and Collaboration**). Users can grab a microphone from the tool menu and speak into it. Their words will be converted to text shown on a notification note and users can place the note near the object to be annotated (see Fig. 3d).

4 ENERGY DATA SCENARIO

We present a use case scenario to show how our system can support all of the design criteria and the entire analysis process with embodied provenance. In our scenario, Susan and Charlie, as data analysts, would like to analyse historical energy consumption data of five key buildings from 2019-2020 on a university campus. The video for this scenario is included in the video figure of this paper.

Susan first focuses on the exploration of data in 2019 (before the outbreak of Covid-19). She investigates the monthly data in 2019 and compares the data in July and December by creating relevant tabletops. She found energy consumption in July is much higher than that in December and identified a peak value in week 4 after a further investigation of the data in July. She speculates that this may because a new semester starts in week 4 of July. She then uses the Note-Down-Mic to annotate a note containing this finding and her inference to the table, for sharing with other analysts (**C4. Support Annotation**) (see Fig. 4a). She further explores the data in week 4 of July and compares the difference of energy consumption in weekdays and weekends by dragging out more tabletops.

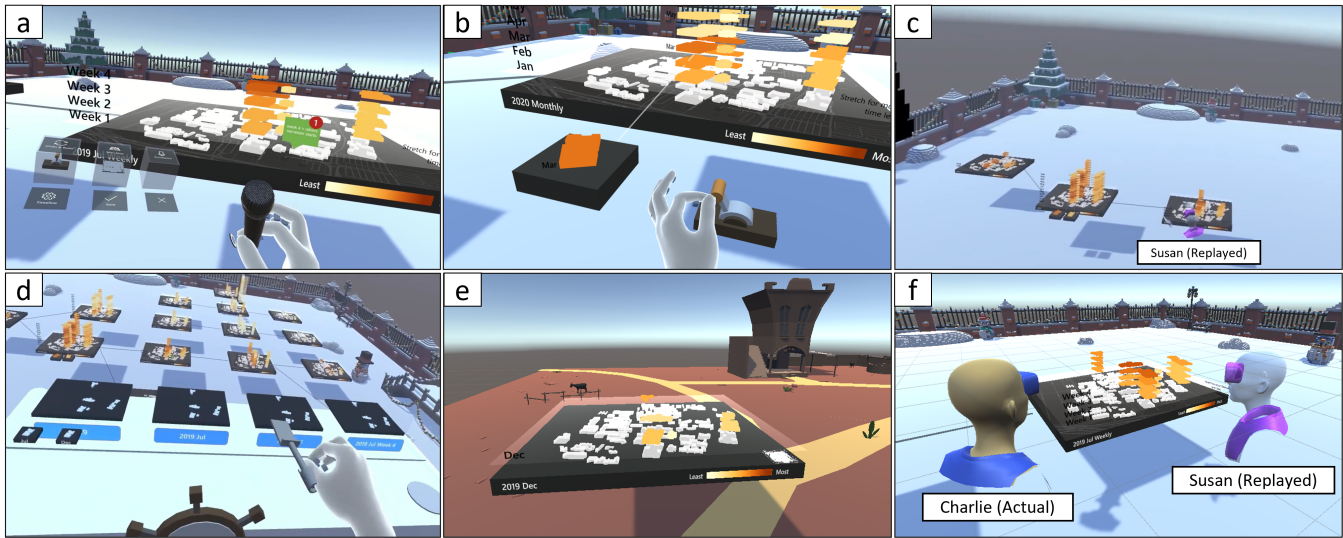


Figure 4: Use case scenario. a) Grab the microphone to add annotation for insights, b) Pull the lever to left to undo from blocks for weekly view to a block for one month, c) Replay of previous analysis conducted by Susan for her own investigation (from the provenance timeline view), d) Copy a specific tabletop using the spatula and e) paste it into a new environment for further investigation, f) Replay of previous analysis conducted by Susan for Charlie's exploration (from the interactive analysis view)

Susan next moves to the exploration of data in 2020 and would like to look at the data in April for a building used for teaching purposes when a lockdown started in April 2020. However, she mistakenly extracts the data in March for that building and further investigates the data for a more detailed time level. After realising this mistake, she uses the Do-It-Lever to undo the actions (**C2. Support Action Recovery**) (see Fig. 4b), removing the detailed time level view.

After a long session of exploration and analysis, she has created a large hierarchy of 15 tabletops along with multiple pedestals for data of certain buildings. She would like to refer back to an analysis step she did for July 2019. She can only remember that there is a snowman in the environment in front of the tabletop but forgets exactly which tabletop it was and the specific analysis she did. So she uses the Replay-It feature to view the playback of the previous analysis process and successfully finds that specific step (**C1. Support Memory Recovery**) (see Fig. 4c).

Now she would like to focus on the tabletop for December 2019 to see if she can find any interesting patterns. However, she doesn't want to modify the existing tabletops in the current environment, so she copies the tabletop using the Copy-It-Spatula and pastes it to a new environment for further investigation (**C3. Support Replication and Reproducibility**) (see Fig. 4d and e).

After a week, Susan goes back to the environments she created to review her analysis. She can easily understand her overall strategy by seeing how the tables showing different time scales and locations are arranged and linked together. She can easily refresh her memory about the steps she took with the help of the annotations and the replay. She then conducts some further analysis and later invites her colleague Charlie to review the analysis and confirm her findings. Charlie enters Susan's environments and quickly understands the

previous analysis done by Susan via revisiting those steps using the replay feature (see Fig. 4f) and investigating the annotations. He discovers an interesting pattern overlooked in the analysis by Susan, so he copies the relevant branch and pastes it to a new environment to further investigate whether the pattern continues across a later time frame (**C5. Support Sharing and Collaboration**).

5 CONCLUSION

In this paper, we proposed a concept of embodied provenance, the use of three-dimensional space and embodiment in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We highlighted a set of design criteria from prior work on analytic provenance and identified key properties for embodied provenance. We presented a prototype system and a use case scenario to demonstrate the concept with multiple data views, spatial environments and novel physical interactions. We plan to conduct a user study to evaluate our prototype for supporting embodied provenance and complete the implementation of the system based on the feedback. We would also work on refining our proposed conceptual framework for embodied provenance. We hope our contributions assist the future design of immersive analytic tools for embodied provenance that allow analytic provenance data to play a stronger role in the sensemaking process.

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